

A per-basin Kalman filter as the reference forecast for data-driven prediction of GRACE terrestrial water storage

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Abstract. Data-driven forecasts of GRACE and GRACE-FO terrestrial water storage anomalies are scored, when a reference forecast is used at all, against climatology or persistence of the last observation. Neither allows for the observation noise in every monthly basin value. We propose a per-basin Kalman filter with an autoregressive state and additive observation noise as the reference forecast for deseasonalized basin storage. It has three parameters per basin, removes noise before extrapolating, and stays defined through months without an observation. On 234 basins over five test folds issued from 2019 to 2026, the filter beats the stronger of two damped-persistence variants by 5.0 % at a lead of one month, 8.8 % at two months, and 2.5 to 5.6 % at leads three to six. Fixing the noise variance at zero removes the whole margin. Scored against the filter, the strongest own-basin model in standardized units is a linear ridge over a flat twelve-month window of filtered state and ERA5-Land forcing, worth 7.6 % at lead one; every sequence model we trained loses to it, significantly at leads two and three against a row-matched LSTM ensemble. Against a published deep-learning product that uses no lagged storage, on two mascon solutions, the filter wins at lead one, wins at lead two only against the product’s non-seasonal component, ties for one or two leads, and loses from lead three to five on, where the product’s forcing history carries it. The lead-one win belongs to the whole reference class: damped persistence also beats the product there, and the filter’s own contribution is its margin over damped persistence, 5.0 % on CSR and indistinguishable from zero on JPL. Skill reported over persistence on this target should be re-read against the filter.

1 Introduction

The GRACE and GRACE-FO missions have measured monthly changes in terrestrial water storage (TWS) since 2002, integrating groundwater, soil moisture, surface water, snow, and ice into one observable that no in situ network can replicate (Tapley et al., 2004; Landerer et al., 2020). Anomalies of this quantity (TWSA) carry months of memory, which makes them both a target and a source of predictability: TWSA forecasts support drought early warning and water management (Li et al., 2026; Arsenault et al., 2020), flood-potential assessment (Reager et al., 2014), and the bridging of the two- to three-month latency of GRACE-FO data releases (Mo et al., 2025).

A growing literature forecasts TWSA directly from data. Neural networks have been trained on GRACE and hydrometeorological series at basin scale (Ahmed et al., 2019), at grid scale globally (Li et al., 2020, 2024; Li and Kusche, 2026), for Europe with subsequent assimilation (Li et al., 2025), and for latency filling (Mo et al., 2025); statistical time-series methods

have been applied per basin (Ahi and Cekim, 2021); benchmark predictability has been quantified at decadal scale (Zhu et al., 2019); and physically based systems forecast TWS with seasonal meteorological forcing (Li et al., 2026; Getirana et al., 2020). Few of these studies score against any reference forecast at all: skill is typically reported as correlation or Nash–Sutcliffe efficiency against the observations, whose implicit reference is the observed mean (a climatology), or against another forecast system, as Li and Kusche (2026) do against SEAS5; we found none that scores against damped persistence. The implicit logic is that a model shown to have skill has extracted predictive structure beyond simple memory. The one study that examines memory directly, the FEWS NET land data assimilation hindcast of Li et al. (2026), concludes that initial conditions dominate sub-seasonal to seasonal TWS skill, but it measures that dominance with a lag-autocorrelation diagnostic rather than with a reference forecast.

This paper argues that persistence is the wrong reference for this target, and proposes a replacement. Basin-average TWSA is a state variable observed through a noisy instrument, with observation errors that are large for small and arid basins (Save et al., 2016; Landerer et al., 2020). A persistence forecast, damped or not, starts from the last observed month, which we call the anchor observation, and propagates its observation noise into every forecast. The reference we propose is a per-basin Kalman filter (Kalman, 1960) with a first-order autoregressive (AR(1)) state and additive observation noise. It separates the signal from the noise before extrapolating, has three parameters per basin, and needs no data beyond the storage series itself. It also differs from persistence in a way this study does not test: because the filter propagates its state through any month without an observation, it is defined across the eleven-month gap between the end of GRACE in 2017 and the start of GRACE-FO in 2018, where persistence has no anchor. No forecast scored here is issued across that gap (Sect. 3.2).

The same benchmark-hygiene argument has been made in sea-ice forecasting, where Niraula and Goessling (2021) showed that damped-persistence references that “do not exploit observations optimally” flatter dynamical forecast systems, and built a stronger reference from spatial damping. We make the analogous move for satellite-observed TWSA through signal and noise separation. Within the GRACE literature, Kalman machinery is established but serves other purposes: ensemble Kalman filters assimilate TWSA into land-surface models (Springer et al., 2026), state-space smoothers clean the gravity-field record (Kankanige et al., 2026), and integrated postprocessing separates the non-seasonal signal from noise in the monthly solutions (Zhang et al., 2025). None of these uses a univariate filter on the finished basin series as a forecast reference, which is the role it plays here; our filter assimilates nothing into any model. The precedent for proposing a reference forecast for TWS is Zhu et al. (2019), who benchmarked decadal TWS predictability with an elasticity framework; we do the same at monthly leads with a filter. Most directly, Nie et al. (2025) showed on 515 basins that deep architectures do not beat linear regression for TWS prediction when autoregressive information is excluded from the feature set by design. Their design has no lagged storage, so it has neither a persistence nor a filter reference; we show the complementary result that when lagged storage is included, how one filters it decides the model comparison.

The contribution of this paper is one claim with three consequences. A per-basin Kalman filter with an AR(1) state and additive observation noise is the reference forecast that data-driven forecasts of deseasonalized GRACE basin water storage should be scored against, because it removes observation noise before extrapolating. On 234 basins over five test folds it beats the stronger of two damped-persistence variants by 5.0 % at lead 1, 8.8 % at lead 2, and 2.5 to 5.6 % at leads 3 to 6, and

refitting it with the noise variance fixed at zero removes the entire margin. Scored against it, a linear ridge correction over a flat twelve-month history of filtered state and ERA5-Land forcing is the strongest own-basin model in standardized units, and every sequence model we trained loses to it. And against a published deep-learning product that takes no lagged storage, on two mascon solutions, the filter wins at lead 1, as damped persistence does, ties for one or two leads, and loses from lead 3 or 4 on CSR and from lead 4 or 5 on JPL, where that product’s forcing history carries it.

Section 2 describes the data, Sect. 3 the target, the folds, the reference forecasts, the corrections built on the filter, the comparison protocol, and the statistics. Section 4 presents the ladder of reference forecasts defined in Sect. 3.3, the ablations that explain the filter’s margin, the flat-window ridge and the sequence models, the comparison with the published product on both mascon solutions, the sensitivity of the rankings to basin weighting, and the experiments that were run and dropped. Section 5 draws the implications and states the limitations.

2 Data

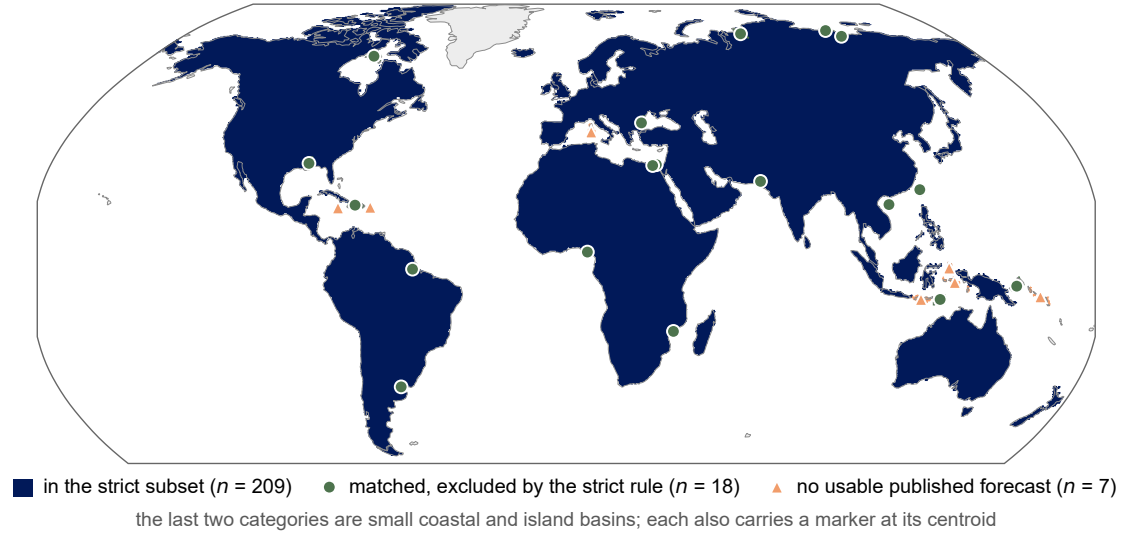
2.1 GRACE and GRACE-FO mascon solutions

The primary target is the CSR RL0603M mascon solution (Save et al., 2016) on a 0.25° grid, April 2002 through May 2026, in centimeters of equivalent water height; the distributed file is titled RL0603M and reports product version RL06.2. Each monthly solution is assigned to its official calendar month using the product’s documented missing-month list, with the solution’s data span asserted to overlap its assigned month; midpoint binning would mislabel the November 2011 and May 2015 solutions, whose arcs are centered on 31 October 2011 and 27 April 2015. The second target is the JPL RL06.3Mv04 mascon solution with the coastline resolution improvement filter applied, sampled at 0.5° , April 2002 through June 2026, with the product’s scale factors applied and its official missing-month metadata used. Both products have a native resolution of roughly 3° equal-area mascons and both are released with correlated-error filtering already applied. The two are independent processings of the same satellite observations, which is what makes the second one useful: a property of the reference forecast that depends on the processing rather than on the observations should not survive the change of product.

2.2 Basins

Basins are defined by a HydroSHEDS-based Level-3 basin and mascon partition (Lehner and Grill, 2013). Each basin series is the cosine-latitude area-weighted mean of its mask cells. From the 284 basins in the partition we exclude 46 ice-sheet basins (Antarctica, Greenland) and 4 water bodies, leaving 234 basins on the CSR product. On the JPL product six small-island masks have no finite scaled cells, so 228 basins are kept there. Figure 1 maps the basins and marks the subsets used in the comparison with the published product (Sect. 3.6).

(a) CSR mascons: the 234-basin sample and the strict comparison subset



(b) JPL mascons: the strict comparison subset

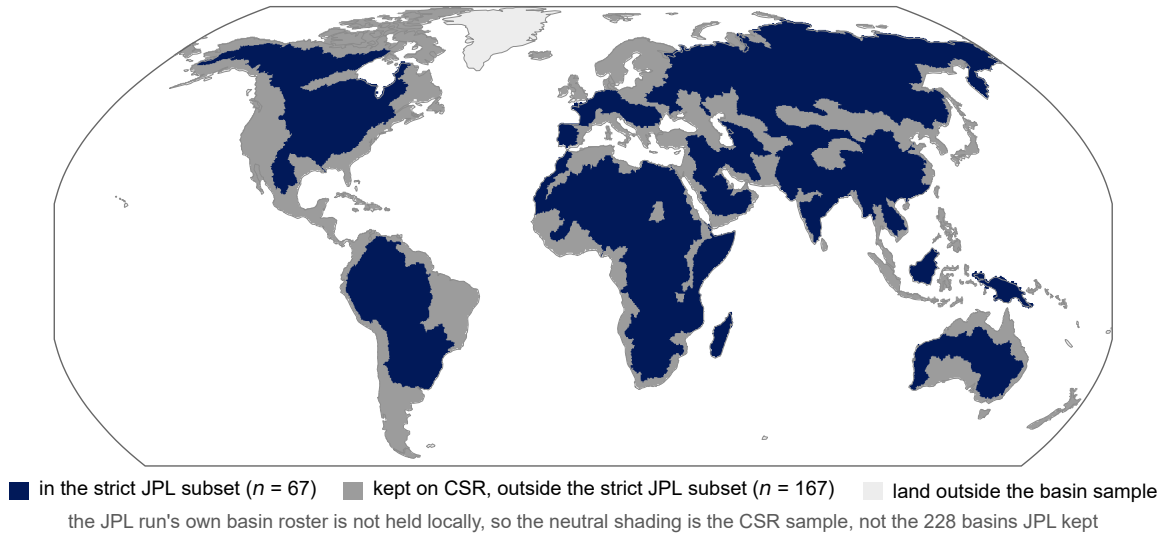


Figure 1. The basin sample. (a) The 234 basins kept on the CSR product, colored by membership in the strict comparison subset of Sect. 3.6: 209 basins enter the comparison with the published product, 18 are matched to that product but fail the full-containment rule, and 7 have no usable cell of the product at all. (b) The JPL product: the 67 basins that pass the rule there are colored, and every other basin of the CSR sample is shaded neutral, because the archived JPL comparison files carry the strict roster only. Panel (b) is therefore not the JPL counterpart of the three-way split in (a). Land outside the basin sample is pale grey in both panels.

2.3 ERA5-Land forcing

90 We use 11 ERA5-Land variables (Muñoz-Sabater et al., 2021) at 0.5° , regridded server-side from 0.1° , January 2001 through May 2026: total precipitation, evaporation, total, surface and subsurface runoff, volumetric soil water in four layers, 2-m temperature, and snow depth water equivalent. Accumulated fluxes are converted from mean daily rates to monthly totals, evaporation is sign-flipped so that positive means moisture loss, temperature is converted to degrees Celsius, and snow depth to centimeters of water equivalent. Basin aggregation uses the same cosine-latitude weighted means as the storage aggregation, mapping each 0.25° mask cell to its nearest 0.5° ERA5 cell. Because ERA5-Land is defined over land only, coastal and island basins aggregate their forcing from the land fraction of their mask: of the 234 basins, 79 draw less than 90 % of their mask weight from valid ERA5 land cells and 13 less than 50 % (minimum 24.5 %). No basin is excluded for this. Twelve monthly climate indices from NOAA PSL enter one reference model only (Sect. 3.3); two of them, AMM and PDO, end before the test window and are dropped from it.

100 2.4 The published forecast product

For the external comparison we use the public GRACE-FCast hindcast of Li and Kusche (2026), downloaded from PANGAEA (Li, 2025). It is released as three products built on the JPL, CSR and GSFC mascons; we use the CSR member against our CSR target and the JPL member against our JPL target, so that each comparison shares a gravity solution. The CSR member holds monthly 1° global forecasts at leads 1 to 12 months, initialized December 2009 through May 2024 (174 initializations); the JPL member holds 173 initializations through April 2024. Their long short-term memory network forecasts the interannual and sub-seasonal components of TWSA from antecedent hydrometeorological and oceanographic predictors with correlation-based spatial predictor shifting; by design it takes no lagged storage as input, and its seasonal and trend components are extrapolated from the previous year and added back (Li and Kusche, 2026). We evaluate both the full product and the non-seasonal product (Sect. 3.6). Three facts qualify the comparison. The skill figures reported by Li and Kusche (2026) are for the average of the three members, and the CSR member is the one their training-residual diagnostic ranks least certain (3.0 against 2.8 cm), so our numbers characterize the members we score, not their headline product. Their JPL member was trained against JPL RL06.1, whereas our JPL target is RL06.3Mv04, so that comparison crosses a release. And their training fills the 2017–2018 mission gap with the reconstruction of Yin et al. (2023), which is hindsight information our models never see.

3 Methods

115 3.1 Target construction and deseasonalization

For each basin and each evaluation fold we fit, on the training window only, a climatology consisting of a linear trend plus annual and semiannual harmonics (six coefficients, ordinary least squares). The frozen coefficients deseasonalize both training and test data; the residual is divided by its training-window standard deviation. The forecast target is this deseasonalized, per-basin-standardized residual, which holds the interannual anomalies that drought and flood applications need. All pooled

120 metrics are computed in this standardized space, so every basin counts equally; Sect. 4.5 and Appendix C report the same forecasts in centimeters, where large basins dominate.

3.2 Folds and the matched sample

We use five expanding-window folds. Their issue windows begin in June 2019, November 2020, April 2022, September 2023 and February 2025, and jointly cover June 2019 through April 2026 at lead 1, with targets reaching May 2026. Fold membership
125 is defined on forecast issue dates: within each fold, every climatology, filter parameter, model coefficient and standardization moment is fit only on observations before the fold’s first issue date, and every test forecast is issued at or after that date. A row issued before the fold start but targeting a month after it belongs to neither set. Splitting on target dates instead would let a lead- h forecast use transforms fit up to $h - 1$ months past its issue date, which is why the issue-date convention matters. Earlier issue windows are unusable: the 2017–2018 gap destroys the lag windows needed to forecast across it, so no scored forecast
130 in this study is issued across the gap. The same rows are scored for every model at a given lead, so the sample is $n = 19,422$, 19,188, 18,954, 18,720, 18,486 and 18,252 basin-months at leads 1 to 6 on the CSR product, and 19,152 at lead 1 on the JPL product, whose record runs one month longer, to June 2026, so its 228 basins carry 84 lead-1 issue months against 83 on CSR. Issue time is defined relative to the last available observation, so the release latency of GRACE-FO is not modeled; all results are retrospective potential skill.

135 3.3 The reference ladder

We score every model against an ordered set of reference forecasts of increasing strength, which we call the reference ladder. Its members are: zero-anomaly climatology; persistence, the last observed anomaly carried forward unchanged; damped persistence in two variants, one that multiplies the last anomaly by a per-basin AR(1) coefficient raised to the lead and one that regresses the lead- h target on the last anomaly, of which we always score against the stronger variant at each lead; a pooled
140 ridge on own-basin lagged anomalies (penalty 1, lags 0, 1, 2, 5 and 11 months, one model for all basins); the same ridge with the climate-index lags of Sect. 2.3 added; and a per-basin ridge on the same lags, refit separately for each basin. The two damped variants are not interchangeable. On CSR the AR(1) variant is the stronger at lead 1 and the regression variant at leads 2 to 6, and the weaker one trails by 1.3 % at lead 1 and by 0.6 to 12.0 % at leads 2 to 6; on JPL the AR(1) variant is the stronger at leads 1 and 2 and the regression variant at leads 3 to 6. Scoring against the weaker one would flatter every other row, so we
145 do not, and every table and figure caption names the variant it uses; the one ablation file scored against the regression variant at every lead is identified as such in Sect. 4.2.

3.4 The Kalman reference

The proposed reference models each basin’s deseasonalized residual y_t as an AR(1) latent signal x_t observed with additive white noise:

$$x_t = \rho x_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, q), \quad (1)$$

$$y_t = x_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, r). \quad (2)$$

The three parameters (ρ, q, r) are estimated per basin and fold by maximum likelihood on training months only, using L-BFGS-B from several starting points. The filter runs forward through the training and test months, updating its state estimate $x_{t|t}$ at each observed month with the Kalman gain, which depends on ρ and on the ratio q/r . At a month with no observation, including all eleven months of the mission gap, the filter skips the update and propagates its state and its state uncertainty forward; nothing is interpolated or filled. This is what makes the reference defined across the gap, where persistence has no anchor; since no scored forecast is issued across the gap (Sect. 3.2), it is a design property of the reference, not a tested one. The lead- h forecast issued at month t is

$$\hat{y}_{t+h|t} = \rho^h x_{t|t}, \quad (3)$$

which is damped persistence applied to the filtered state instead of to the raw observation. We call this forecast the Kalman reference. It is not a data-assimilation scheme: it assimilates nothing into any model and estimates no state but the basin’s own scalar signal.

Two variants of the filter are used to explain its margin. The *noise-free filter* refits the identical model by the identical likelihood with r fixed at zero. That forces the gain to one, so the state tracks the raw observation and the forecast degenerates to damped persistence with a maximum-likelihood ρ ; the difference between the two filters is the value of the noise term alone. The *mission-split filter* gives the observation equation two noise variances, one for months before January 2018 and one for GRACE-FO months from then on, so the parameter vector is $(\rho, q, r_{\text{GRACE}}, r_{\text{FO}})$, with everything else unchanged. A fold whose training window holds fewer than 12 GRACE-FO months cannot separate the second variance from q and is refit with the single-variance model and flagged.

One caveat governs how the noise term should be read. The split between signal variance q and observation variance r is a modeling choice, not a measurement: fast hydrology that an AR(1) state cannot carry forward, and decomposition error, are absorbed into r alongside instrument noise. In a minority of fits the likelihood pushes r to its lower bound: 259 of the 1170 basin-fold fits land there. So r is not, basin by basin, a clean estimate of instrument noise, and we describe what the filter removes as observation noise in the model’s sense, not the instrument’s.

3.5 Corrections built on the filter

Every own-basin model beyond the reference predicts the error of the Kalman reference, and its forecast is the Kalman forecast plus that predicted error. Four are linear. The *state ridge* regresses the error on the current filtered state alone. The *flat-12 ridge*

regresses it on the last twelve months of the filtered state, flattened into one feature vector. The *ERA5 flat-12 ridge* adds the twelve months of all 11 ERA5-Land anomaly series to that vector, giving a window of 12 by 12 features. The *ERA5 lag ridge* adds instead the ERA5 anomalies at lags 0, 1 and 2 months to the current filtered state. All four are pooled across basins with penalty 1 on train-standardized features. Three are sequence models or nonlinear heads on the same information. The *LSTM* is a shared-encoder long short-term memory network (Hochreiter and Schmidhuber, 1997) over the same twelve-month window of filtered state and the 11 ERA5 channels, with a time-ordered split holding out the last 15 % of training months for early stopping; two seeds, reported as a seed ensemble. The *flat-history MLP* is a multilayer perceptron (hidden layers 64 and 32, early stopping) on the ERA5 flat-12 ridge’s feature vector; two seeds. The *residual MLP* is a two-stage model, the ERA5 lag ridge followed by an MLP of the same size trained on the first stage’s training residual; three seeds. Two *neighbor corrections*, tested and dropped in Sect. 4.6, add to the state ridge one other basin’s filtered state propagated to the target month, that basin chosen by correlation or by distance. We call the five models that are also reported in centimeters (Sect. 4.5 and Appendix C) the *retained models*: the stronger damped-persistence variant, the Kalman reference, the state ridge, the flat-12 ridge and the ERA5 flat-12 ridge. Table 1 collects the names, each used for exactly one model throughout; Appendix A maps them to the identifiers in the result files.

3.6 Comparison protocol for the published product

The hindcast of Li (2025) is aggregated to our basin geometry by exact 0.25° to 1° cell-containment weights. Their forecasts are moved into each fold’s deseasonalized standardized space using our train-window climatology plus a per-fold, per-basin, per-lead mean offset estimated on pre-test data only. The full product is scored after subtracting our climatology, so any difference between their seasonal and trend convention and ours is absorbed there; the non-seasonal product, which is their interannual plus sub-seasonal forecast, is scored in the same space and removes the advantage their monthly re-extrapolation of the seasonal cycle and trend would otherwise carry. The matched sample is the identical set of basin-months for every model at every lead. On CSR that is 60 test months: folds 1 to 3 in full, 9 of the 17 issue months of fold 4, and none of fold 5, because the product’s final initialization, May 2024, falls inside fold 4; on JPL, whose final initialization is one month earlier, it is 59.

Two basin populations are reported. The matched population keeps every basin with at least one usable cell of the product, which is 227 of the 234 basins on CSR and 227 of 228 on JPL. The strict population applies one rule to both products: a basin enters only if it fully contains every 0.25° mask cell of at least one native mascon of the product being scored and all sixteen cells of at least one finite 1° cell of the published forecast. This is a literal containment test, not an area or coverage proxy. On CSR, 229 basins contain a full native mascon, 211 contain a full forecast cell, and 209 satisfy both; on JPL, whose native mascons are coarser on the sampled grid, 67 do. The strict population is the one we report, and the reason is visible in the matched one on JPL: there, the published product’s standardized error at lead 1 reads 5.47 on the matched population against 0.93 on the strict one. The inflation is not a defect in the product or in the aggregation. It is our fold standardization by the residual standard deviation of our own product on basins that the coarser forecast grid only partly covers, and the remedy is the sample restriction. We do not quote the matched-population numbers as statements about the product’s accuracy.

Table 1. Model names used in the paper, with the defining subsection. Appendix A maps each name to its identifier in the archived result files.

Name	Definition
Climatology	Zero-anomaly forecast (Sect. 3.3).
Persistence	Last observed anomaly (Sect. 3.3).
Damped persistence	Last anomaly damped toward zero; AR(1) and regression variants, always scored against the stronger at each lead (Sect. 3.3).
Pooled lag ridge	Ridge on own-basin lagged anomalies, one model for all basins; the index variant adds climate-index lags (Sect. 3.3).
Per-basin lag ridge	The pooled lag ridge’s features, refit separately for each basin (Sect. 3.3).
Kalman reference	The proposed reference: damped persistence of the filtered state from the per-basin AR(1) plus observation-noise model (Sect. 3.4).
Noise-free filter	The same model refit with the observation variance fixed at zero (Sect. 3.4).
Mission-split filter	The same model with separate observation variances for GRACE and GRACE-FO months (Sect. 3.4).
State ridge	Kalman reference plus a ridge correction on the current filtered state (Sect. 3.5).
Flat-12 ridge	Kalman reference plus a ridge correction on the flattened 12-month window of filtered state (Sect. 3.5).
ERA5 flat-12 ridge	The flat-12 ridge with the 12-month window of the 11 ERA5-Land anomalies added (Sect. 3.5).
ERA5 lag ridge	Kalman reference plus a ridge on the current filtered state and ERA5-Land anomalies at lags 0 to 2 (Sect. 3.5).
LSTM	Shared-encoder LSTM over the 12-month window of filtered state and ERA5-Land channels; two-seed ensemble (Sect. 3.5).
Flat-history MLP	Multilayer perceptron on the ERA5 flat-12 ridge’s features; two seeds (Sect. 3.5).
Residual MLP	ERA5 lag ridge followed by an MLP on its training residual; three seeds (Sect. 3.5).
Neighbor corrections	Kalman reference plus a ridge on the own state and one other basin’s propagated state, selected by correlation or by distance (Sect. 3.5; tested and dropped in Sect. 4.6).
Published product (GRACE-FCast), full and non-seasonal	The published product of Li and Kusche (2026), with and without its extrapolated seasonal and trend terms (Sects. 2.4 and 3.6).

3.7 Statistics

The primary metric is pooled root-mean-square error (RMSE) in standardized units. We report it as the skill score $1 - \text{MSE}_A / \text{MSE}_B$ of model A against a stated reference B , in percent. Pooled comparisons use the Diebold–Mariano test (Diebold

and Mariano, 1995) on the monthly cross-basin mean loss differential, with the small-sample correction of Harvey et al. (1997) and a Newey–West variance at maximum lag $\max(h-1, 1)$ (Newey and West, 1987). Confidence intervals use a moving-block bootstrap (Künsch, 1989) of the monthly losses with block length $\max(3, h)$ and 2000 draws; the block covers the lag- $(h-1)$ dependence of overlapping h -step forecasts. Per-basin significance uses per-basin Diebold–Mariano statistics on the scored test months with Benjamini–Hochberg false-discovery-rate control at $q = 0.10$ (Benjamini and Hochberg, 1995), following the field-significance reasoning of Wilks (2016); helped and hurt basins are always counted separately. Every skill in this paper is stated with our model first, so a positive value always means our forecast has the lower error. The archived JPL comparison tables list the published product first; a skill s stored that way is converted to our convention as $1 - 1/(1 - s)$, the reciprocal of the mean-squared-error ratio, and not by a change of sign.

4 Results

4.1 The reference ladder

Table 2 and Fig. 2 give the ladder on the matched rows. The Kalman reference beats the stronger damped-persistence variant at every lead, by +4.98, +8.79, +5.62, +3.07, +2.55 and +3.63% at leads 1 to 6, with Diebold–Mariano p values of 3.5×10^{-12} , 6.3×10^{-19} , 5.7×10^{-19} , 2.6×10^{-11} , 7.9×10^{-7} and 2.0×10^{-9} . The margin is largest at lead 2 and the p values do not run in order of lead. The bootstrap interval on the lead-1 margin is +3.9 to +6.1%.

The filter also beats the per-basin lag ridge, the strongest conventional model in the ladder at short leads, at every lead: +1.99, +3.04, +1.43, +1.04, +2.17 and +3.92%, with $p \leq 0.013$ at five leads. At lead 4 the test does not reject ($p = 0.078$), although the bootstrap interval there excludes zero (+0.02 to +2.20%); the Diebold–Mariano test is the stated criterion, so we count five leads, not six. Against the pooled lag ridge the picture is different and worth stating exactly: the filter is ahead at leads 1, 2, 3 and 6 and behind at leads 4 and 5, and the difference is significant at leads 1 and 2 only (+3.86%, $p = 9.1 \times 10^{-6}$; +4.21%, $p = 1.2 \times 10^{-11}$; then +0.57, −0.59, −0.64 and +1.17%, all with $p \geq 0.19$). A pooled ridge with five lag coefficients shared across 234 basins is a competent long-lead extrapolator; what it cannot do is filter the anchor observation, and at leads 1 and 2 that is what matters.

Three features of Table 2 deserve comment. First, the gap between raw and damped persistence, 16 to 23%, is the familiar cost of not damping; the further 5.0 to 8.8% at leads 1 and 2 from damping the filtered state instead of the observation is of the same order as many published margins of learned models over persistence. Second, the learned models without noise separation sit between the two persistence variants and the filter: the per-basin lag ridge, with five coefficients per basin, never reaches the three-parameter filter, and by lead 6 it has fallen below damped persistence. Third, the reference is nearly free. It adds no data requirements and a few lines of state-space code, and it is the strongest forecast of the reference class, at three parameters per basin. The state ridge, which adds one regression coefficient on the filtered state, is stronger at leads 3 to 6, by +0.44, +2.36, +2.66 and +1.59% ($p \leq 1.6 \times 10^{-13}$), and it is behind at leads 1 and 2 (−0.17 and −1.01%). Both are the same filtered state read two ways; neither uses any information persistence does not have.

Table 2. The reference ladder: pooled skill (%) against the stronger damped-persistence variant at each lead, on matched rows (234 basins, five folds; n as in Sect. 3.2). Positive is better. The reference is the AR(1) variant at lead 1 and the regression variant at leads 2 to 6. The best value in each column is in bold. Appendix B gives the standardized RMSE and the test p value for every row.

Model	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$	$h=6$
Climatology	−125.6	−70.7	−58.9	−51.4	−42.7	−33.6
Persistence	−20.7	−22.7	−21.4	−19.5	−16.0	−15.8
Damped persistence, weaker variant	−1.3	−0.6	−6.4	−11.8	−12.0	−10.0
Damped persistence, stronger variant	0.0	0.0	0.0	0.0	0.0	0.0
Pooled lag ridge	+1.2	+4.8	+5.1	+3.6	+3.2	+2.5
Pooled lag ridge, index variant	+1.0	+4.5	+5.1	+3.7	+3.2	+2.4
Per-basin lag ridge	+3.1	+5.9	+4.3	+2.1	+0.4	−0.3
Kalman reference	+5.0	+8.8	+5.6	+3.1	+2.5	+3.6
State ridge	+4.8	+7.9	+6.0	+5.4	+5.1	+5.2
Flat-12 ridge	+5.7	+10.1	+7.3	+4.8	+4.0	+4.2
ERA5 flat-12 ridge	+12.2	+13.7	+9.6	+6.2	+4.6	+4.5

Skills are computed on matched rows from the archived per-prediction files; the source file for every table and figure is listed in the archive manifest described in the code and data availability statement.

On the second mascon solution, the JPL product with 228 basins and the same folds, the margin holds at leads 2 to 6 and not at lead 1. The Kalman reference beats the stronger damped variant by +6.35, +4.64, +1.55, +2.36 and +4.74 % at leads 2 to 6 ($p \leq 0.017$), and by +0.56 % at lead 1 with $p = 0.60$, a tie. Against the per-basin lag ridge it is likewise behind at lead 1 (−0.66 %, $p = 0.46$) and ahead at leads 2 to 6 (+2.11 to +6.67 %, $p \leq 0.026$). The lead-1 tie on JPL is consistent with what
250 the ablations of the next section say the filter is doing: JPL’s coarser mascons with the coastline filter deliver a basin series with less month-to-month noise for the filter to remove, and the standardized persistence error on JPL is correspondingly smaller (damped-persistence RMSE 0.67 at lead 1 in standardized units, against 1.08 on CSR). Where there is less noise to remove, the filter has less to gain at the lead where noise matters most.

4.2 Why the filter wins

255 Two ablations attribute the margin (Fig. 3). The noise-free filter, the same model refit with $r = 0$, loses to the full filter at every lead, by 5.35 % at lead 1, rising to 12.65 % at lead 5 and 12.27 % at lead 6 ($p \leq 7.2 \times 10^{-14}$ at every lead). Scored against the regression-damped variant, which the ablation file uses at every lead, the noise-free filter is +0.90 % ahead at lead 1 ($p = 1.6 \times 10^{-5}$) and behind at leads 2 to 6, by 0.93 to 11.57 %; against the AR(1) variant, the stronger one at lead 1, it is 0.40 % behind (standardized RMSE 1.0783 against 1.0761). The lead-1 margin of Table 2 therefore decomposes into a +5.35 %
260 noise term, the Kalman reference over the noise-free filter, and a 0.40 % deficit of the noise-free filter’s maximum-likelihood ρ against the AR(1) variant’s; the two mean-squared-error ratios multiply to the +4.98 % of Table 2. Figure 3a, drawn against

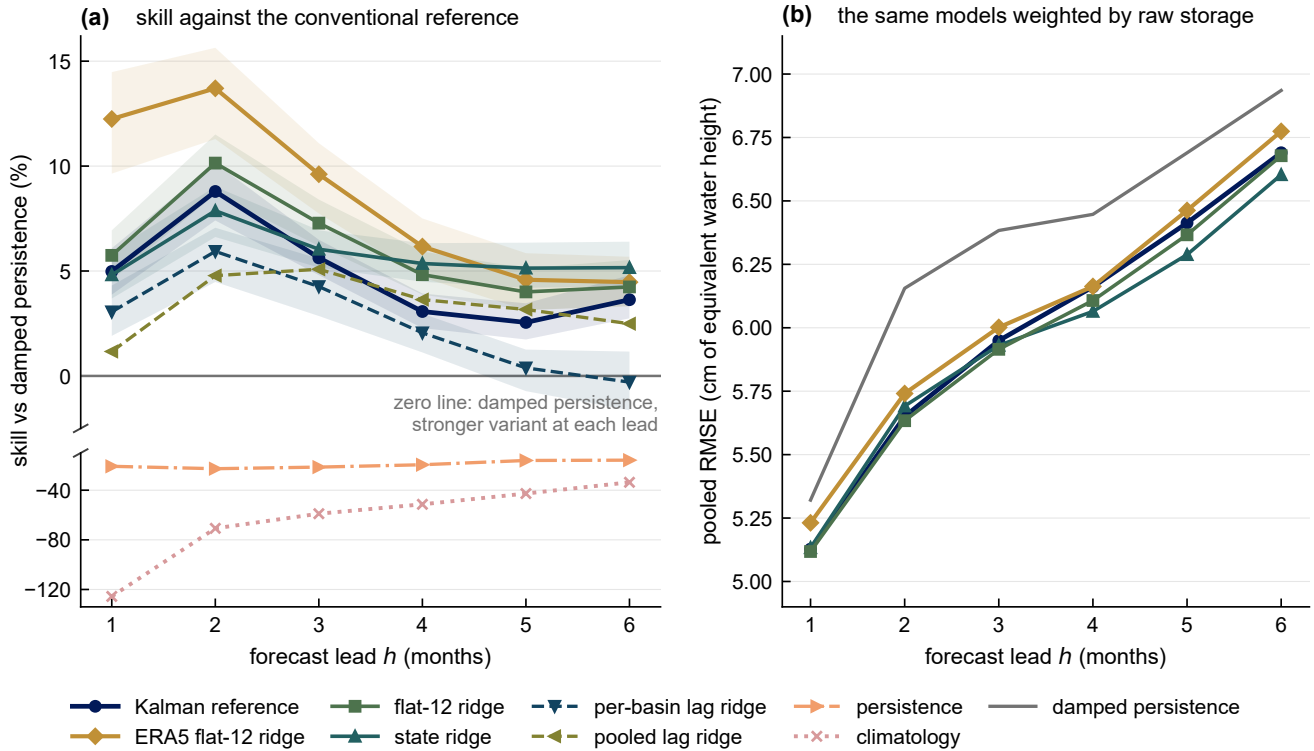


Figure 2. The reference ladder. (a) Pooled skill (%) against the stronger damped-persistence variant, the AR(1) variant at lead 1 and the regression variant at leads 2 to 6, for the ERA5 flat-12 ridge, the Kalman reference, the flat-12 ridge, the state ridge, the per-basin lag ridge, the pooled lag ridge, persistence and climatology, the last two on a broken axis. Moving-block bootstrap 95 % confidence intervals are drawn where a contrast against that variant exists in the archived files: for the ERA5 flat-12 ridge, the Kalman reference, the flat-12 ridge, the state ridge and the per-basin lag ridge. The gap between the zero line and the Kalman reference is the gain from filtering the anchor observation before damping it. (b) Pooled RMSE in centimeters of equivalent water height by lead for the five retained models of Sect. 3.5: damped persistence (stronger variant), the Kalman reference, the state ridge, the flat-12 ridge and the ERA5 flat-12 ridge. In this weighting the largest basins dominate and the ranking changes (Sect. 4.5).

the regression variant at every lead, reads +6.21 % for the Kalman reference at lead 1 for the same reason. The AR(1) structure without the noise term is therefore worth nothing beyond damped persistence; the margin is the act of filtering the observation before extrapolating. Because r absorbs whatever an AR(1) state cannot carry forward (Sect. 3.4), this ablation attributes the margin to filtering within the model class, not to a particular physical noise source.

The second ablation answers the referee's question that the two missions raise: does the margin depend on treating GRACE and GRACE-FO as one instrument? It does not, and the opposite change hurts. The mission-split filter, with a separate observation variance for GRACE-FO months, is worse than the single-variance filter at every lead: -1.84% at lead 1 ($p = 1.7 \times 10^{-7}$), then -0.80 , -0.65 , -0.57 , -0.53 and -0.38% ($p \leq 0.0060$). Per basin at lead 1, after false-discovery-rate control, the split

Table 3. Corrections built on the filter: pooled skill (%) against the Kalman reference at each lead on matched rows (upper value), with the Diebold–Mariano p value below it. Positive means the correction improves on the reference; the best value in each column is in bold.

Model	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$	$h=6$
State ridge	−0.17 1.8×10^{-6}	−1.01 1.9×10^{-11}	+0.44 1.1×10^{-14}	+2.36 1.6×10^{-13}	+2.66 5.9×10^{-14}	+1.59 2.3×10^{-17}
Flat-12 ridge	+0.81 0.0035	+1.48 4.9×10^{-6}	+1.76 2.9×10^{-10}	+1.81 7.5×10^{-10}	+1.49 3.4×10^{-7}	+0.63 0.015
ERA5 flat-12 ridge	+7.65 1.1×10^{-8}	+5.39 1.1×10^{-8}	+4.22 3.0×10^{-8}	+3.19 5.2×10^{-6}	+2.09 9.0×10^{-4}	+0.87 0.12

The state ridge’s p values are two-sided tests of a difference that is negative at leads 1 and 2 and positive at leads 3 to 6.

270 helps 0 basins and hurts 0 of the 234 tested; the loss is diffuse, not concentrated. The first fold’s training window ends in May 2019 and holds fewer than 12 GRACE-FO months, so all 234 of its basin-fold fits fall back to the single variance, which is the fallback rule of Sect. 3.4 operating as designed. Every scored month is a GRACE-FO month, and a variance fit to under six years of them is estimated worse than one fit to the whole record. We keep the single-variance filter.

4.3 The flat-12 ridge and the sequence models

275 What beats the filter is the question the reference exists to sharpen, and Table 3 answers it for the own-basin models. The flat-12 ridge, which reads twelve months of the filtered state instead of one, adds +0.63 to +1.81 % over the filter at every lead ($p \leq 0.015$). The ERA5 flat-12 ridge adds the twelve-month forcing window and gains +7.65% over the filter at lead 1 (95 % interval +4.7 to +10.1 %), +5.39, +4.22, +3.19 and +2.09 % at leads 2 to 5 ($p \leq 9.0 \times 10^{-4}$), and +0.87 % at lead 6, which is not significant ($p = 0.12$). Over damped persistence that is +12.24 % at lead 1 and +13.71 % at lead 2, the largest margins
280 in the study. The ERA5 flat-12 ridge is the strongest own-basin model in standardized units at leads 1 to 4 only. At leads 5 and 6 the state ridge, with one coefficient on the current filtered state, is marginally better: the ERA5 flat-12 ridge scores −0.58 % against it at lead 5 and −0.73 % at lead 6, which in standardized RMSE is 1.3804 against 1.3764 and 1.4421 against 1.4369. The forcing window pays for itself through lead 4 and not beyond.

Where the ERA5 window helps is not uniform across basins. Figure 4 maps the per-basin lead-1 skill of the ERA5 flat-
285 12 ridge over the filter and plots it against each basin’s training-window storage standard deviation in centimeters. The gain concentrates in basins with small storage variance, where a standardized unit is a small physical amount and the forcing history explains a larger share of it. This is the property that makes the ranking sensitive to basin weighting (Sect. 4.5).

The sequence models are the other half of this section. Every sequence model we trained loses to the ERA5 flat-12 ridge on the same twelve months of history (Table 4 and Fig. 5). At lead 1 the standardized RMSE of the ERA5 flat-12 ridge is 1.0081,
290 against 1.0132 and 1.0188 for the two LSTM seeds, 1.0235 and 1.0291 for the two flat-history MLP seeds, 1.0254 for the ERA5 lag ridge, and 1.0490 for the filter. The flat ridge is lowest at leads 2 and 3 as well. The flat-history MLP, which sees the

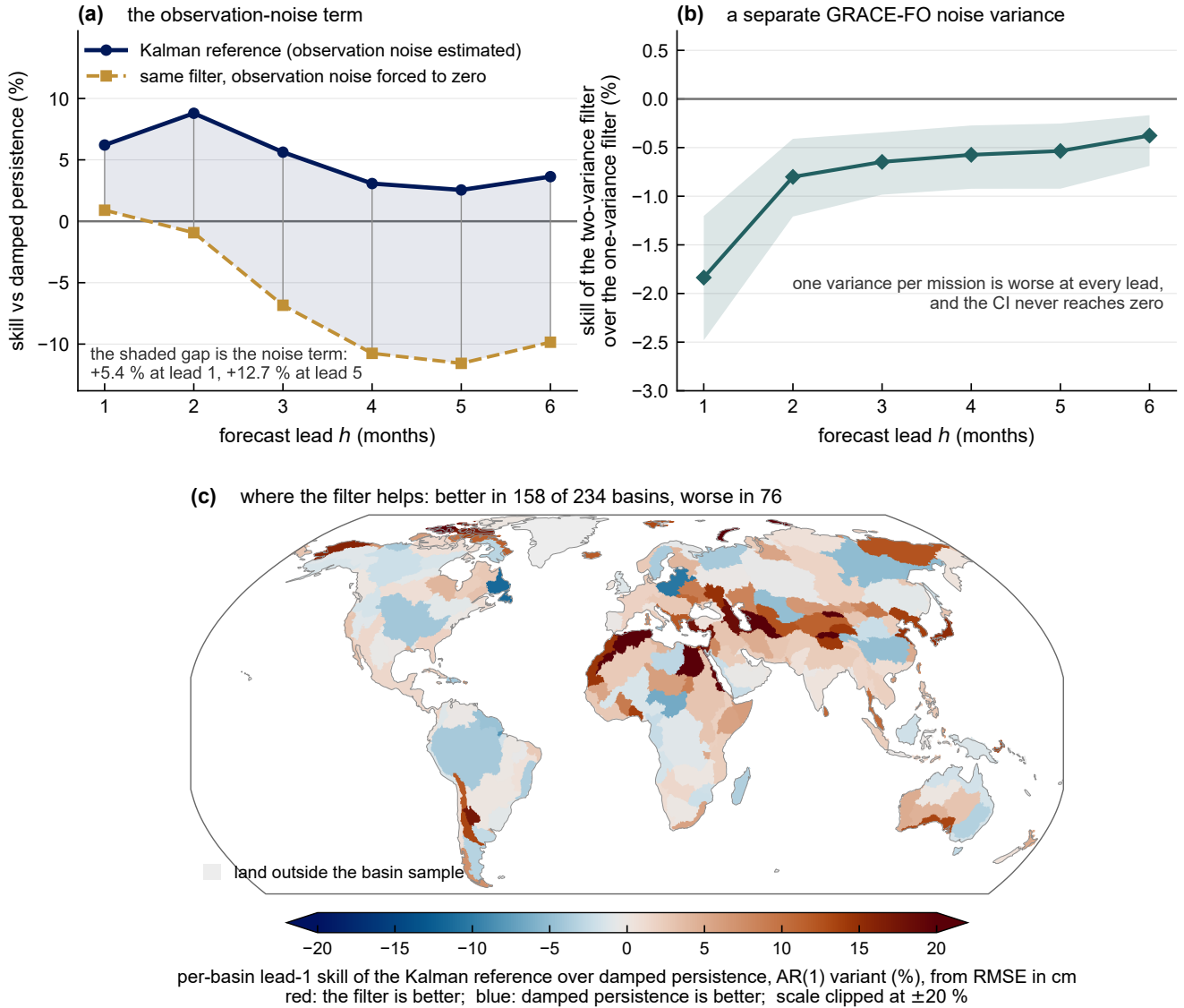


Figure 3. Why the filter wins. (a) Skill (%) against the regression damped-persistence variant at every lead for the Kalman reference and for the noise-free filter, the same model refit with $r = 0$. The shading is the gap between the two curves, which is the value of the noise term, and carries no confidence interval. Because the reference here is the regression variant at every lead, the Kalman reference reads +6.21 % at lead 1 rather than the +4.98 % of Table 2, which uses the stronger AR(1) variant at that lead. (b) The mission-split filter against the single-variance Kalman reference by lead, with moving-block bootstrap 95 % confidence intervals; negative means the split hurts. (c) Per-basin lead-1 skill of the Kalman reference over the AR(1) damped-persistence variant on the 234 basins, computed from RMSE in centimeters; the color scale is clipped at ± 20 %, and land outside the basin sample is pale grey.

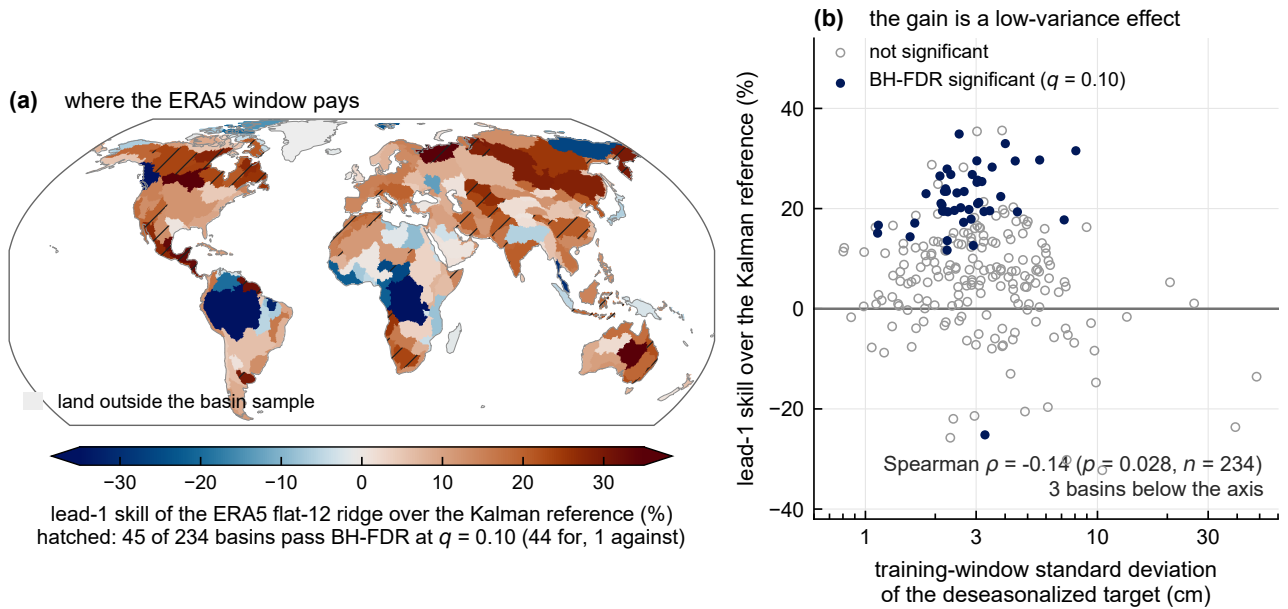


Figure 4. Where the forcing window helps. (a) Per-basin lead-1 skill (%) of the ERA5 flat-12 ridge over the Kalman reference on the 234 basins; basins significant under Benjamini–Hochberg false-discovery-rate control at $q = 0.10$ are hatched, and land outside the basin sample is pale grey. (b) The same skill against each basin’s training-window storage standard deviation in centimeters of equivalent water height; three basins fall below the lower axis limit of -42% and are counted on the panel. The gain concentrates in low-variance basins.

identical feature vector as the flat ridge, is significantly worse than it in every seed at every lead ($p \leq 2.3 \times 10^{-6}$). The residual MLP, the two-stage model of Sect. 3.5, reaches 1.0145 to 1.0156 at lead 1 across its three seeds, also behind the flat ridge. The LSTM holds out its last 15 % of training months for early stopping, so its comparison with a ridge fit on all training rows is not row-matched. Refitting the ERA5 flat-12 ridge on the LSTM’s exact training rows settles it: against the two-seed LSTM ensemble the row-matched ridge is ahead by $+1.02\%$ at lead 1 ($p = 0.094$), $+2.76\%$ at lead 2 ($p = 2.7 \times 10^{-10}$) and $+2.27\%$ at lead 3 ($p = 4.3 \times 10^{-9}$), and it is indistinguishable from the ridge fit on all rows (-0.17 , $+0.04$ and $+0.19\%$, all $p \geq 0.19$). Under row matching the lead-1 margin is not significant on its own; at leads 2 and 3 it is. Twelve months of forcing carry real information, a linear reader extracts it, and the sequence architectures pay a premium for re-learning what the flattening gives them directly.

4.4 Against the published product on CSR and JPL

Table 5 and Fig. 6 give the comparison with GRACE-FCast on the strict subset. On CSR, over 209 basins and 60 months, 12,540 rows per lead, the Kalman reference alone beats the full product at lead 1 by $+16.4\%$ (95 % interval $+4.9$ to $+26.6\%$, $p = 2.2 \times 10^{-3}$), ties at lead 2 (-6.0% , $p = 0.23$), and loses from lead 3 on, by 16.3 to 35.4% ($p \leq 1.4 \times 10^{-3}$). Against the

Table 4. Sequence models against the flat ridge: standardized pooled RMSE at leads 1 to 3 on matched rows (n as in Sect. 3.2). Lower is better. Seeds are listed separately; the LSTM ensemble is scored in the text through the row-matched comparison.

Model	$h=1$	$h=2$	$h=3$
Kalman reference	1.0490	1.1954	1.2734
ERA5 lag ridge	1.0254	1.1930	1.2685
Flat-12 ridge	1.0448	1.1865	1.2622
ERA5 flat-12 ridge	1.0081	1.1627	1.2462
LSTM, seed 0 / seed 1	1.0132 / 1.0188	1.1758 / 1.1851	1.2578 / 1.2630
Flat-history MLP, seed 0 / seed 1	1.0235 / 1.0291	1.1783 / 1.1935	1.2602 / 1.2799
Residual MLP, seeds 0 / 1 / 2	1.0148 / 1.0156 / 1.0145	1.1760 / 1.1788 / 1.1773	1.2635 / 1.2530 / 1.2540

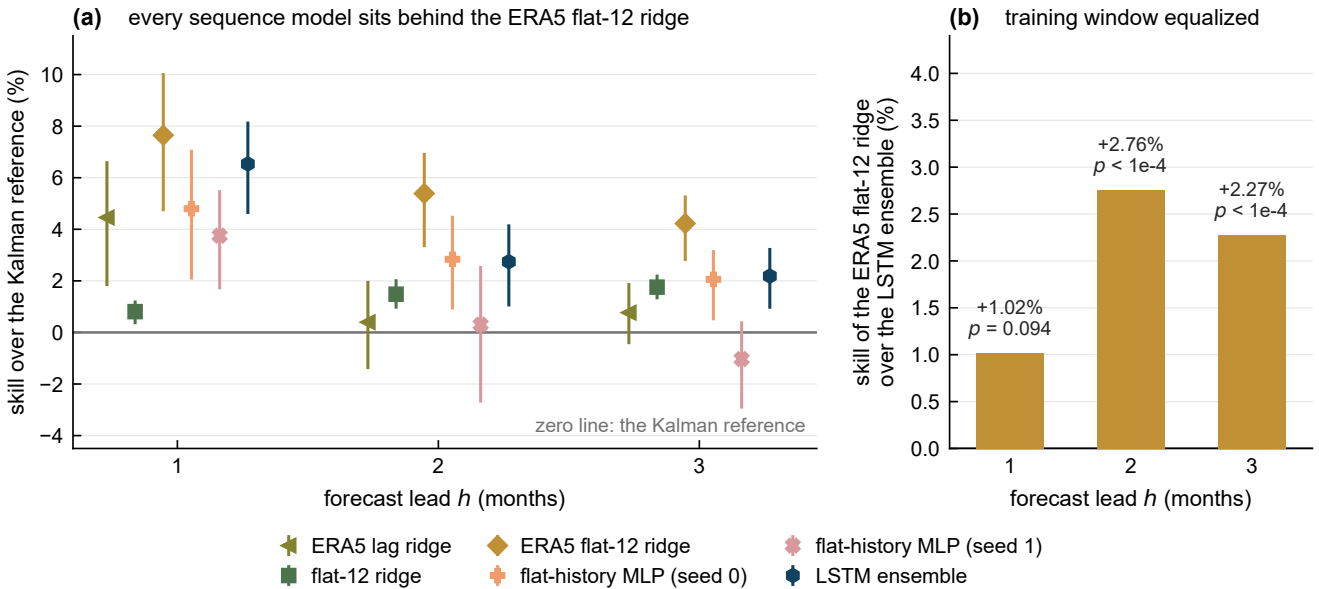


Figure 5. Sequence models against the flat ridge. (a) Skill (%) over the Kalman reference at leads 1 to 3 for the ERA5 lag ridge, the flat-12 ridge, the ERA5 flat-12 ridge, the two seeds of the flat-history MLP and the two-seed LSTM ensemble, with moving-block bootstrap 95 % confidence intervals. The residual MLP has no archived predictions and appears in Table 4 only. (b) Skill of the ERA5 flat-12 ridge over the LSTM ensemble when the ridge is refit on the LSTM’s exact training rows, with the Diebold–Mariano p value. Every sequence model sits below the linear reader of the same twelve-month history.

305 non-seasonal product, which removes the seasonal and trend re-extrapolation, the reference wins at leads 1 and 2 (+28.4%, $p = 2.7 \times 10^{-10}$; +7.5%, $p = 0.018$), ties at lead 3 (−4.0%, $p = 0.13$) and loses from lead 4. The ERA5 flat-12 ridge, our strongest own-basin model at these leads, moves the crossing later by one lead in each pairing: against the full product it

Table 5. The comparison with GRACE-FCast on the strict subset, CSR product: 209 basins, 60 months, 12,540 rows per lead. Skill (%) of our forecast over theirs (upper value), so positive means ours has the lower error, as in every table of this paper, with the Diebold–Mariano p value below it. The last two rows count the basins, of 209, in which the published product has the lower error at leads 1 and 6.

Pair	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$	$h=6$
Kalman vs. full	+16.4	−6.0	−16.3	−23.2	−28.9	−35.4
	0.0022	0.23	0.0014	3.9×10^{-5}	1.7×10^{-6}	1.7×10^{-8}
Kalman vs. non-seasonal	+28.4	+7.5	−4.0	−13.4	−20.3	−27.1
	2.7×10^{-10}	0.018	0.13	4.3×10^{-6}	7.8×10^{-9}	5.9×10^{-10}
ERA5 flat-12 ridge vs. full	+21.5	−1.8	−12.1	−19.6	−26.4	−34.6
	4.0×10^{-4}	0.74	0.032	0.0017	7.3×10^{-5}	4.8×10^{-7}
ERA5 flat-12 ridge vs. non-seasonal	+32.8	+11.2	−0.2	−10.1	−18.0	−26.4
	2.4×10^{-10}	0.0021	0.94	0.0032	6.7×10^{-6}	3.3×10^{-8}
Basins, full product beats Kalman	98					157
Basins, full product beats ERA5 flat-12	87					162

wins lead 1 (+21.5%, $p = 4.0 \times 10^{-4}$), ties lead 2 (−1.8%, $p = 0.74$) and loses from lead 3 (−12.1%, $p = 0.032$, widening to −34.6%); against the non-seasonal product it wins leads 1 and 2 (+32.8%, interval +24.0 to +40.8%, $p = 2.4 \times 10^{-10}$; 310 +11.2%, interval +3.5 to +18.5%, $p = 2.1 \times 10^{-3}$), ties lead 3 (−0.2%, $p = 0.94$) and loses from lead 4 (−10.1%, $p = 3.2 \times 10^{-3}$, widening to −26.4%).

Basin by basin the picture is a near-even split at lead 1 that tips as the lead grows: the full product has the lower error in 98 of the 209 basins against the Kalman reference at lead 1 and in 157 at lead 6, and in 87 against the ERA5 flat-12 ridge at lead 1 and 162 at lead 6. The pooled lead-1 margin is therefore magnitude in the basins where we win, not breadth. The 315 reason the reference wins at all is structural, and it is a property of the reference class, not of the filter: their network takes no lagged storage, so at lead 1, where the most recent observation is the strongest predictor available, any forecast anchored on that observation has information their product lacks. On the same subset damped persistence itself beats the full product at lead 1 by +11.9% ($p = 0.029$) and loses to it at leads 2 to 6 by 16.9 to 54.3%. The filter’s own contribution at lead 1 is its margin over damped persistence, the +4.98% of Table 2, not the +16.4% of Table 5. Nor should the product’s lead-1 deficit 320 be over-read: damped persistence is not available to their intended user, since the two- to three-month release latency their product exists to bridge withholds the anchor observation.

On JPL, over 67 basins and 59 months (3,953 rows per lead), only the Kalman reference has been scored: the JPL analysis was completed before the flat-12 models were built, and they will be added. The archived JPL tables list the published product first; the values here are converted to our convention (Sect. 3.7). Against the full product the reference scores +43.7% at lead 325 1 ($p = 7.1 \times 10^{-14}$), +9.8% at lead 2 ($p = 0.087$, a tie), −6.2% at lead 3 ($p = 0.31$, a tie), and −19.1, −32.1 and −41.9% at leads 4 to 6 ($p \leq 1.7 \times 10^{-3}$). Against the non-seasonal product it scores +52.9% at lead 1 ($p = 1.3 \times 10^{-13}$), so its mean

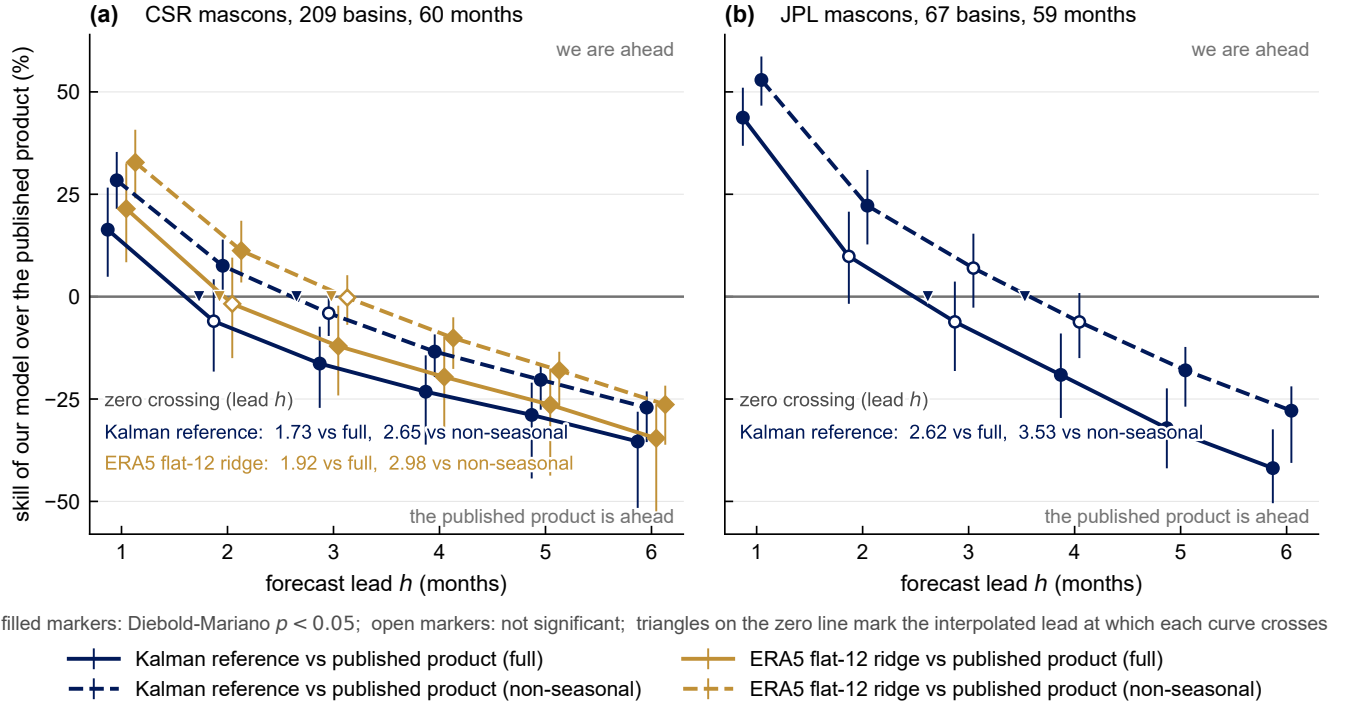


Figure 6. The comparison with the published product on the strict basin subset. (a) CSR, 209 basins and 60 months: skill (%) of the Kalman reference and of the ERA5 flat-12 ridge over the published product, full and non-seasonal, by lead, with moving-block bootstrap 95 % confidence intervals; positive means our forecast has the lower error. (b) JPL, 67 basins and 59 months, Kalman reference only; the archived JPL values are stored with the product first and are converted with the reciprocal transform of Sect. 3.7. Filled markers are significant at $p < 0.05$, open markers are not, and the triangles on the zero line mark each curve’s interpolated crossing. Against the full product the reference wins lead 1, ties one lead on CSR and two on JPL, and loses from lead 3 on CSR and lead 4 on JPL; against the non-seasonal product it wins leads 1 and 2, ties one lead on CSR and two on JPL, and loses from lead 4 on CSR and lead 5 on JPL.

squared error there is less than half the product’s, +22.2% at lead 2 ($p = 2.1 \times 10^{-5}$), +7.0% at lead 3 ($p = 0.15$) and −6.2% at lead 4 ($p = 0.16$), both ties, and −18.0 and −27.9% at leads 5 and 6 ($p \leq 1.1 \times 10^{-4}$). The pattern is the CSR one shifted later by a lead: against the full product the reference wins lead 1, ties leads 2 and 3 and loses from lead 4; against the non-seasonal product it wins leads 1 and 2, ties leads 3 and 4 and loses from lead 5. Here too the lead-1 win is a reference-class result. Damped persistence beats the full product on the same rows by +43.2%, nearly the filter’s margin, and on JPL the filter ties damped persistence at lead 1 (+0.56%, $p = 0.60$; Sect. 4.1), so on this product the filter adds nothing at lead 1 that persistence of the anchor observation does not already have. The lead-1 margin over the product is much larger on JPL than on CSR. Two things differ there, and we do not separate them: the strict subset on JPL is a set of 67 large basins with no small ones, and the JPL member of the product was trained on an earlier release than the one we score it against (Sect. 2.4).

4.5 Weighting sensitivity: standardized against raw centimeters

Every ranking above is in standardized units, where each basin counts equally. In pooled raw centimeters, where the largest basins dominate, the ranking among the retained models changes (Fig. 2b and Appendix C). The flat-12 ridge without forcing has the lowest pooled RMSE at leads 1 to 3, 5.118 cm at lead 1 against 5.128 for the filter and 5.231 for the ERA5 flat-12 ridge; the state ridge is lowest at leads 4 to 6 (6.064, 6.288 and 6.603 cm); and the ERA5 flat-12 ridge sits behind the plain filter at every lead (5.231 to 6.774 cm against 5.128 to 6.690). Damped persistence is worst in this weighting too, at 5.321 to 6.935 cm. The deposited file also carries a stacked LSTM ensemble from an extended run that is not a retained model and supports no claim in this paper; a reader of the file will find it at 5.144, 5.743, 5.935, 6.019, 6.226 and 6.510 cm at leads 1 to 6, below the state ridge at leads 4 to 6. The filter’s margin over persistence is therefore not a property of the weighting, but the forcing window’s gain is: it lives in the low-variance basins of Fig. 4, which the standardized metric counts in full and the centimeter metric barely counts. A ranking of models on this target has to say which weighting it refers to, and “strongest own-basin model” in this paper means lowest pooled standardized error.

4.6 Tested and dropped: neighboring basins

Spatial and graph-based TWSA methods assume that one basin’s history helps forecast another’s future: Li and Kusche (2026) shift predictors across space, Arzoumanidis et al. (2026) build a proximity and correlation basin graph for reconstruction, and Steidl and Zhu (2025) forecast with a hierarchical basin graph. We tested the simplest version of that assumption: whether one other basin’s filtered state, propagated to the target time and added as a feature to the state ridge, improves the forecast. On CSR a correlation-selected neighbor was worth +0.31% over the state ridge at lead 1, ahead of all 50 seed-matched random neighbor graphs and all 99 amplitude-adjusted Fourier-transform surrogates (Schreiber and Schmitz, 1996), and worth nothing at leads 2 to 6. On JPL none of it replicates. There every neighbor variant is worse than the state ridge at every lead; the correlation-selected neighbor scores -0.52 to -1.28% and beats 0 of 50 random graphs at every lead, the distance-selected neighbor beats 0 of 50 at leads 1 to 3 and 8, 26 and 9 of 50 at leads 4 to 6, and the real graph beats 0 of 99 surrogates at every lead. A controlled effect that reverses on a second processing of the same observations is not a finding, and we make no claim about it. The experiments remain in the archived code.

5 Discussion

5.1 What the reference means for reported skill

The result is an argument about reference classes, not a new forecasting method. A persistence reference is supposed to represent no information beyond memory. For a noisy observable with a gap in its record it leaves the memory class’s own headroom unused, and on deseasonalized GRACE basin series that headroom is 5.0 to 8.8% at the leads where most data-driven papers claim their skill. The inflation compounds down the ladder: a study scoring against damped persistence concedes this margin; one scoring against raw persistence concedes a further 16 to 23%; one scoring against climatology concedes far

more (Table 2). The forecasting studies cited in Sect. 1 sit at the bottom of that ladder or off it: where a reference exists it is a climatology, often implicit in a Nash–Sutcliffe score, or another forecast system, and none that we examined scores against damped persistence. Skill reported on observed TWSA therefore cannot be read as information beyond what a three-parameter filter extracts. The fix costs three parameters per basin and no data.

The comparison with the published product shows what the fix changes in practice. GRACE-FCast is a serious system, and at leads 4 to 6 every model we built gives up 10 to 35 % of skill to it; its forcing history carries information that no own-basin model has. At lead 1 it loses to any forecast anchored on the observation it does not use, damped persistence included, and at lead 2 its non-seasonal component loses to the filter. Neither statement is visible in a full-signal correlation score. Li and Kusche (2026) document the same target sensitivity from the other side: they report correlation above 0.8 over about 72 % of global land at lead 1 for the full signal but only 20 % once the seasonal cycle and trend are removed, and attribute the difference to the extrapolated components directly. Most of the correlation in a full-signal GRACE forecast score is carried by terms that are extrapolated, not predicted, and Appendix C shows the same thing for our own forecasts. The deseasonalized target with a filter reference is the one on which forecast systems for this variable can be compared.

Two scope notes. The argument applies to observed TWSA. Targets that are themselves model output, reconstructions or gap fills, such as the HydroGlobe target of Nie et al. (2025) or the products of Humphrey and Gudmundsson (2019), Yin et al. (2023) and Mo et al. (2022), contain no satellite observation noise to filter, so neither the inflation nor its fix transfers to them. And the filter is not data assimilation (Springer et al., 2026): nothing is assimilated into any model, and the only state estimated is the basin’s own scalar signal. The closest upstream machinery we know of is the postprocessing of Zhang et al. (2025), which separates the non-seasonal GRACE signal from noise in the monthly solutions to improve extraction of the record; ours points the separated state forward, using it to forecast and to score forecasts.

5.2 Relation to prior benchmarks

Three prior results bear directly on this one. Nie et al. (2025) found on 515 basins that deep architectures do not beat linear regression for TWS prediction. Their feature set excludes lagged storage by design, so their comparison has no persistence reference and no filter reference; it asks how well forcing predicts storage, not how well the storage record predicts itself. Our result is the complement. With lagged storage included, the model comparison is decided by how the storage is filtered, and once it is filtered a linear reader of the forcing history beats every sequence model we trained (Sect. 4.3). The two results agree that linear models are the benchmark to beat on this target; ours adds that the benchmark’s own reference should be a filter. Zhu et al. (2019) proposed a benchmark for decadal TWS predictability from an elasticity framework, on the argument that a field without an agreed reference cannot compare its forecasts. The same argument applies at monthly leads, and the filter is the reference we propose. And Niraula and Goessling (2021) showed for the sea-ice edge that a damped-persistence reference which does not use the observations optimally flatters the systems scored against it; the filter is the analogous correction for a noisy scalar observable. The FEWS NET hindcast of Li et al. (2026) concluded that initial conditions dominate sub-seasonal to seasonal TWS skill; the filter turns that qualitative statement into a reference forecast that a model has to beat.

400 5.3 The crossing and the initial-conditions versus forcing partition

The seasonal hydrologic forecasting literature settled a structurally identical question with physically based models: skill at short leads comes from initial conditions and skill at long leads from meteorological forcing, with a crossover between them (Wood and Lettenmaier, 2008; Shukla and Lettenmaier, 2011; Koster et al., 2010; Yossef et al., 2013; Pechlivanidis et al., 2020). Those results were established for streamflow and soil moisture in ensemble streamflow prediction experiments. The comparison of Sect. 4.4 locates the same shape on satellite-observed storage between two real forecast systems, one built entirely on the observed state and one built entirely on forcing. On both mascon solutions the filter wins at lead 1 against both product variants and at lead 2 against the non-seasonal variant, the two tie for one lead on CSR and two on JPL, and the forcing-driven product wins from lead 3 or 4 on CSR and from lead 4 or 5 on JPL. The short-lead half of that shape belongs to the reference class rather than to the filter, since damped persistence wins at lead 1 as well (Sect. 4.4). We describe the crossing as consistent with the partition rather than as a decomposition of skill into sources. The two systems differ in architecture, training data, gap handling and predictor sets at once, so what is measured is the lead profile of each system, not the effect of any one design choice.

The practical reading is a design map. Below the crossing, invest in the initial condition: three parameters filter it, and none of the architectures we trained improved on a linear reader of the filtered state. Above it, invest in forcing and in the depth of forcing history a model reads: the twelve-month ERA5 window carried our own-basin skill through lead 4 (Sect. 4.3), and a model trained on twelve months of spatially shifted forcing carries it to lead 6 and beyond. A system that hands over from the first to the second at the crossing is the obvious next step, and the reference proposed here is what its short-lead half should be scored against.

5.4 Limitations

1. *No climate forecasts, so long leads are out of reach.* None of our models uses forecast forcing, and the ERA5 window ends at the issue month. The published product's advantage from lead 3 or 4 on CSR and from lead 4 or 5 on JPL is the advantage of a forcing history read with a model trained on it, and the reference we propose has nothing to offer at those leads beyond what the filtered state carries. A hybrid that hands over from the filter to a forcing-driven system at the crossing is the obvious next system; we do not build it here, and the crossing lead would have to be chosen on data we have not used.
2. *The JPL comparison covers the Kalman reference only.* The JPL analysis was completed before the flat-12 models were built; they will be added. Until then the two-product claim applies to the reference alone. The JPL strict subset is 67 large basins, and the JPL member of the published product was trained on an earlier release than we score it against; the size of the lead-1 margin there should not be compared with CSR's.

- 430 3. *The mission gap is handled by construction, not by test.* The filter is defined across the 2017–2018 gap because its predict step runs when an update is skipped, whereas persistence has no anchor there; but no scored forecast in this study is issued across the gap (Sect. 3.2), so this property is stated, not demonstrated.
4. *Standardized weighting favors small basins.* The ranking of corrections above the filter depends on weighting, and the forcing window’s gain is concentrated in low-variance basins that pooled centimeter error barely counts (Sect. 4.5). The
435 filter’s margin over persistence holds in both weightings.
5. *The noise term is not identified as instrument noise.* 259 of 1170 basin-fold fits push r to its lower bound (Sect. 3.4), and the AR(1) plus white-noise decomposition absorbs unmodeled dynamics into r . The ablation attributes the margin to filtering within the model class; what is filtered out is a mixture of instrument noise and fast hydrology whose shares we do not estimate.
- 440 6. *The lead-1 margin does not replicate on JPL.* On the second product the filter ties damped persistence at lead 1 and reproduces the margin at leads 2 to 6 (Sect. 4.1). Our reading, that JPL’s processing leaves less month-to-month noise for the filter to remove, is consistent with the smaller persistence error there but is not tested directly.
7. *Comparison asymmetries with the published product.* Their training uses the gap reconstruction of Yin et al. (2023); their full product re-extrapolates the seasonal cycle and trend monthly, an edge the non-seasonal comparison removes;
445 our lead-1 win is one that damped persistence shares and partly reflects a reference tuned to exactly this target definition; and the matched sample excludes fold 5 and the basins their grid does not cover.
8. *Sample and era.* Two mascon products of one set of observations; five test folds all in the post-2019 climate era; per-basin estimates rest on 83 lead-1 months. Two seeds for the LSTM and the flat-history MLP and three for the residual MLP: seed spread at lead 1 is comparable to the gaps in Table 4, which is why no single seed is reported as a result.

450 6 Conclusions

For basin-average, deseasonalized GRACE water storage, the standard persistence reference is weaker than its own class. A per-basin Kalman filter with an AR(1) state and additive observation noise, fit on the same observations with three parameters, beats the stronger damped-persistence variant by 5.0 % at lead 1, 8.8 % at lead 2 and 2.5 to 5.6 % at leads 3 to 6 on 234 basins over five test folds, and the per-basin lag ridge at five of six leads. Removing the noise term removes the margin; splitting the
455 noise variance by mission makes the filter worse. On a second mascon solution the margin holds at leads 2 to 6 and is a tie at lead 1. The filter is defined across the mission gap by construction; forecasts across the gap are not scored here. We propose the filter as the reference forecast for this task.

Scored against it, the strongest own-basin model in standardized units at leads 1 to 4 is a linear ridge over a flat twelve-month window of filtered state and ERA5-Land forcing, worth 7.6 % over the filter at lead 1; at leads 5 and 6 a single coefficient on
460 the filtered state is marginally better; and every sequence model we trained loses to the flat ridge on the same history. Against

a published deep-learning product that takes no lagged storage, on two mascon solutions, the filter wins at lead 1 against both product variants and at lead 2 against the non-seasonal variant, ties for one lead on CSR and two on JPL, and loses from lead 3 or 4 on CSR and from lead 4 or 5 on JPL, where that product’s forcing history carries it. The lead-1 win is one damped persistence shares; the filter’s own contribution there is its margin over damped persistence. Short leads on this target are a
465 filtering problem; longer leads are a forcing problem; and a forecast’s skill on observed GRACE storage should be stated against the filter, not against persistence of the observation.

Code and data availability. CSR GRACE and GRACE-FO mascon solutions are available from the Center for Space Research (<https://www2.csr.utexas.edu/grace/>) and JPL mascon solutions from NASA PO.DAAC (<https://podaac.jpl.nasa.gov/>). ERA5-Land data are available from the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu/>) (Muñoz-Sabater et al., 2021). Climate indices are available from
470 NOAA PSL (<https://psl.noaa.gov/data/climateindices/>). The GRACE-FCast hindcast is available from PANGAEA (Li, 2025). The analysis code is at <https://github.com/ishaankejriwal/graceseespaper>. The reference forecasts, the per-prediction files and every result table used in this paper, together with a manifest mapping each table and figure to its source file, will be deposited on Zenodo before submission at <https://doi.org/10.5281/zenodo.XXXXXXXX>.

Appendix A: Model names and archived identifiers

475 The archived result files identify each model by the identifier used in the analysis code. Table A1 maps the names of Table 1 to those identifiers, which appear nowhere else in the paper. Seed suffixes (`_s0`, `_s1`, `_s2`) denote single training seeds and `_ens` the seed ensemble.

Appendix B: The full reference ladder

Table B1 gives, for every model in Table 2, the pooled standardized RMSE and the Diebold–Mariano p value against the
480 stronger damped-persistence variant at all six leads.

Appendix C: Conventional metrics in centimeters

Table C1 reports the retained models in the metrics the literature uses: pooled RMSE and the per-basin median RMSE in centimeters of equivalent water height, and the per-basin median correlation and Nash–Sutcliffe efficiency, on the deseasonalized anomaly and on the full signal. The full-signal forecast adds each fold’s frozen train-window climatology to the predicted
485 anomaly. Because forecast and observation share that climatology, the centimeter errors are identical in the two spaces; only correlation and efficiency change, and the change is the share of a full-signal score carried by the extrapolated seasonal cycle and trend. At lead 1, damped persistence alone reaches a median full-signal correlation of 0.86 and a median full-signal efficiency of 0.72, values that would read as strong skill in a full-signal evaluation.

Table A1. Model names used in the paper and the corresponding identifiers in the archived result files.

Paper name	Identifier(s)
Climatology	climatology_zero
Persistence	persistence
Damped persistence, AR(1) and regression variants	damped_persistence_rho, damped_persistence_reg
Pooled lag ridge, and index variant	ridge_own_lags, ridge_own_plus_indices
Per-basin lag ridge	ridge_own_perbasin
Kalman reference	kalman_ar1
Noise-free filter	kalman_r0
Mission-split filter	kalman_mission
State ridge	kalman_own_ridge (ridge_own in the sequence-model files)
Flat-12 ridge	ridge_own_flat12
ERA5 flat-12 ridge	ridge_own_era5_flat12; row-matched refit ridge_own_era5_flat12_train85
ERA5 lag ridge	ridge_own_era5
LSTM	lstm_own_era5_s0, lstm_own_era5_s1, lstm_own_era5_ens
Flat-history MLP	mlp_own_era5_flat12_s0, mlp_own_era5_flat12_s1
Residual MLP	resmlp_own_era5_s0, resmlp_own_era5_s1, resmlp_own_era5_s2
Neighbor corrections, correlation- and distance-selected	kalman_corr_top1, kalman_geo_top1
Published product (GRACE-FCast), full and non-seasonal	li_lstm_full, li_lstm_nonseas
Strict comparison subset	joint_full_cells; matched population all_matched

Author contributions. I.K. designed the study, wrote the analysis code, performed the experiments and audits, and wrote the manuscript.

490 *Competing interests.* The authors declare that they have no conflict of interest.

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Table B1. Pooled standardized RMSE (upper value) and Diebold–Mariano *p* value against the stronger damped-persistence variant (lower value) for every model of Table 2 at leads 1 to 6, on the matched rows of Sect. 3.2. The reference row has no test.

Model	<i>h</i> =1	<i>h</i> =2	<i>h</i> =3	<i>h</i> =4	<i>h</i> =5	<i>h</i> =6
Climatology	1.6163	1.6354	1.6525	1.6698	1.6881	1.7055
	6.0×10^{-39}	1.8×10^{-41}	3.3×10^{-36}	1.2×10^{-34}	2.8×10^{-35}	1.7×10^{-34}
Persistence	1.1825	1.3863	1.4442	1.4837	1.5224	1.5877
	1.4×10^{-11}	4.7×10^{-11}	3.7×10^{-11}	1.7×10^{-7}	2.7×10^{-4}	5.9×10^{-4}
Damped persistence, weaker variant	1.0831	1.2552	1.3519	1.4354	1.4953	1.5478
	2.1×10^{-4}	0.13	2.6×10^{-19}	1.3×10^{-23}	2.3×10^{-21}	7.9×10^{-22}
Damped persistence, stronger variant	1.0761	1.2516	1.3108	1.3573	1.4132	1.4754
	–	–	–	–	–	–
Pooled lag ridge	1.0698	1.2214	1.2770	1.3324	1.3906	1.4570
	0.15	2.4×10^{-5}	3.7×10^{-6}	4.3×10^{-5}	0.0026	0.013
Pooled lag ridge, index variant	1.0708	1.2230	1.2766	1.3316	1.3906	1.4578
	0.24	7.0×10^{-5}	3.0×10^{-6}	3.2×10^{-5}	0.0026	0.015
Per-basin lag ridge	1.0596	1.2139	1.2826	1.3433	1.4105	1.4776
	1.6×10^{-5}	2.2×10^{-12}	1.8×10^{-7}	8.5×10^{-5}	0.48	0.70
Kalman reference	1.0490	1.1954	1.2734	1.3363	1.3951	1.4484
	3.5×10^{-12}	6.3×10^{-19}	5.7×10^{-19}	2.6×10^{-11}	7.9×10^{-7}	2.0×10^{-9}
State ridge	1.0499	1.2014	1.2706	1.3204	1.3764	1.4369
	1.5×10^{-11}	5.7×10^{-19}	1.7×10^{-19}	4.9×10^{-18}	6.7×10^{-12}	1.0×10^{-13}
Flat-12 ridge	1.0448	1.1865	1.2622	1.3241	1.3846	1.4438
	5.8×10^{-13}	1.9×10^{-20}	1.6×10^{-19}	1.5×10^{-16}	1.3×10^{-10}	2.7×10^{-11}
ERA5 flat-12 ridge	1.0081	1.1627	1.2462	1.3148	1.3804	1.4421
	3.3×10^{-18}	2.1×10^{-20}	4.0×10^{-17}	3.4×10^{-15}	2.5×10^{-9}	2.5×10^{-12}

495 **References**

Ahi, G. O. and Cekim, H. O.: Long-term temporal prediction of terrestrial water storage changes over global basins using GRACE and limited GRACE-FO data, *Acta Geodaetica et Geophysica*, 56, 321–344, <https://doi.org/10.1007/s40328-021-00338-4>, 2021.

Ahmed, M., Sultan, M., Elbayoumi, T., and Tissot, P.: Forecasting GRACE Data over the African Watersheds Using Artificial Neural Networks, *Remote Sensing*, 11, 1769, <https://doi.org/10.3390/rs11151769>, 2019.

500 Arsenault, K. R., Shukla, S., Hazra, A., Getirana, A., McNally, A., Kumar, S. V., Koster, R. D., Peters-Lidard, C. D., Zaitchik, B. F., Badr, H., Jung, H. C., Narapusetty, B., Navari, M., Wang, S., Mocko, D. M., Funk, C., Harrison, L., Husak, G. J., Adoum, A., Galu, G., Magadzire, T., Roningen, J., Shaw, M., Eylander, J., Bergaoui, K., McDonnell, R. A., and Verdin, J. P.: The NASA Hydrological Forecast System for Food and Water Security Applications, *Bulletin of the American Meteorological Society*, 101, E1007–E1025, <https://doi.org/10.1175/BAMS-D-18-0264.1>, 2020.

- 505 Arzoumanidis, L., Johannsen, L., Middendorf, K., Eicker, A., and Dehbi, Y.: Reconstructing GRACE Terrestrial Water Storage with Spatio-Temporal Graph Neural Networks: An Application to South America, <https://arxiv.org/abs/2606.23833>, 2026.
- Benjamini, Y. and Hochberg, Y.: Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing, *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 57, 289–300, <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>, 1995.
- Diebold, F. X. and Mariano, R. S.: Comparing Predictive Accuracy, *Journal of Business & Economic Statistics*, 13, 253–263, <https://doi.org/10.1080/07350015.1995.10524599>, 1995.
- 510 Getirana, A., Rodell, M., Kumar, S., Beaudoin, H. K., Arsenault, K., Zaitchik, B., Save, H., and Bettadpur, S.: GRACE Improves Seasonal Groundwater Forecast Initialization over the United States, *Journal of Hydrometeorology*, 21, 59–71, <https://doi.org/10.1175/JHM-D-19-0096.1>, 2020.
- Harvey, D., Leybourne, S., and Newbold, P.: Testing the equality of prediction mean squared errors, *International Journal of Forecasting*, 13, 281–291, [https://doi.org/10.1016/S0169-2070\(96\)00719-4](https://doi.org/10.1016/S0169-2070(96)00719-4), 1997.
- 515 Hochreiter, S. and Schmidhuber, J.: Long Short-Term Memory, *Neural Computation*, 9, 1735–1780, <https://doi.org/10.1162/neco.1997.9.8.1735>, 1997.
- Humphrey, V. and Gudmundsson, L.: GRACE-REC: a reconstruction of climate-driven water storage changes over the last century, *Earth System Science Data*, 11, 1153–1170, <https://doi.org/10.5194/essd-11-1153-2019>, 2019.
- 520 Kalman, R. E.: A New Approach to Linear Filtering and Prediction Problems, *Journal of Basic Engineering*, 82, 35–45, <https://doi.org/10.1115/1.3662552>, 1960.
- Kankanige, D., Vishwakarma, B. D., Liu, Y. Y., and Sharma, A.: Can a Persistence-Based Prediction of GRACE Gravity Field Coefficients Reduce Uncertainties at Finer Temporal Scales?, *Earth and Space Science*, 13, <https://doi.org/10.1029/2025EA004554>, 2026.
- Koster, R. D., Mahanama, S. P. P., Yamada, T. J., Balsamo, G., Berg, A. A., Boisserie, M., Dirmeyer, P. A., Doblas-Reyes, F. J., Drewitt, G., Gordon, C. T., Guo, Z., Jeong, J.-H., Lawrence, D. M., Lee, W.-S., Li, Z., Luo, L., Malyshev, S., Merryfield, W. J., Seneviratne, S. I., Stanelle, T., van den Hurk, B. J. J. M., Vitart, F., and Wood, E. F.: Contribution of land surface initialization to subseasonal forecast skill: First results from a multi-model experiment, *Geophysical Research Letters*, 37, <https://doi.org/10.1029/2009GL041677>, 2010.
- Künsch, H. R.: The Jackknife and the Bootstrap for General Stationary Observations, *The Annals of Statistics*, 17, 1217–1241, <https://doi.org/10.1214/aos/1176347265>, 1989.
- 530 Landerer, F. W., Flechtner, F. M., Save, H., Webb, F. H., Bandikova, T., Bertiger, W. I., Bettadpur, S. V., Byun, S. H., Dahle, C., Dobslaw, H., Fahnestock, E., Harvey, N., Kang, Z., Kruizinga, G. L. H., Loomis, B. D., McCullough, C., Murböck, M., Nagel, P., Paik, M., Pie, N., Poole, S., Strelakov, D., Tamisiea, M. E., Wang, F., Watkins, M. M., Wen, H.-Y., Wiese, D. N., and Yuan, D.-N.: Extending the Global Mass Change Data Record: GRACE Follow-On Instrument and Science Data Performance, *Geophysical Research Letters*, 47, <https://doi.org/10.1029/2020GL088306>, 2020.
- 535 Lehner, B. and Grill, G.: Global river hydrography and network routing: baseline data and new approaches to study the world’s large river systems, *Hydrological Processes*, 27, 2171–2186, <https://doi.org/10.1002/hyp.9740>, 2013.
- Li, B., Hazra, A., McNally, A., Slinski, K., Shukla, S., and Anderson, W.: Skills in sub-seasonal to seasonal terrestrial water storage forecasting: insights from the FEWS NET land data assimilation system, *Hydrology and Earth System Sciences*, 30, 1097–1115, <https://doi.org/10.5194/hess-30-1097-2026>, 2026.
- 540 Li, F.: A Global Dataset for Seasonal to Annual Forecasts of GRACE-like Gridded Terrestrial Water Storage (2010–2024), <https://doi.org/10.1594/PANGAEA.973113>, 2025.

- Li, F. and Kusche, J.: Observation-Driven Forecast of Global Terrestrial Water Storage and Evaluation for 2010–2024, *Water Resources Research*, 62, <https://doi.org/10.1029/2025WR041710>, 2026.
- Li, F., Kusche, J., Rietbroek, R., Wang, Z., Forootan, E., Schulze, K., and Lück, C.: Comparison of Data-Driven Techniques to Reconstruct (1992–2002) and Predict (2017–2018) GRACE-Like Gridded Total Water Storage Changes Using Climate Inputs, *Water Resources Research*, 56, <https://doi.org/10.1029/2019WR026551>, 2020.
- Li, F., Kusche, J., Sneeuw, N., Siebert, S., Gerdener, H., Wang, Z., Chao, N., Chen, G., and Tian, K.: Forecasting Next Year’s Global Land Water Storage Using GRACE Data, *Geophysical Research Letters*, 51, <https://doi.org/10.1029/2024GL109101>, 2024.
- Li, F., Springer, A., Kusche, J., Gutknecht, B. D., and Ewerdwalbesloh, Y.: Reanalysis and Forecasting of Total Water Storage and Hydrological States by Combining Machine Learning With CLM Model Simulations and GRACE Data Assimilation, *Water Resources Research*, 61, <https://doi.org/10.1029/2024WR037926>, 2025.
- Mo, S., Zhong, Y., Forootan, E., Mehrnegar, N., Yin, X., Wu, J., Feng, W., and Shi, X.: Bayesian convolutional neural networks for predicting the terrestrial water storage anomalies during GRACE and GRACE-FO gap, *Journal of Hydrology*, 604, 127–244, <https://doi.org/10.1016/j.jhydrol.2021.127244>, 2022.
- Mo, S., Schumacher, M., van Dijk, A. I. J. M., Shi, X., Wu, J., and Forootan, E.: Near-Real-Time Monitoring of Global Terrestrial Water Storage Anomalies and Hydrological Droughts, *Geophysical Research Letters*, 52, <https://doi.org/10.1029/2024GL112677>, 2025.
- Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D. G., Piles, M., Rodríguez-Fernández, N. J., Zsoter, E., Buontempo, C., and Thépaut, J.-N.: ERA5-Land: a state-of-the-art global reanalysis dataset for land applications, *Earth System Science Data*, 13, 4349–4383, <https://doi.org/10.5194/essd-13-4349-2021>, 2021.
- Newey, W. K. and West, K. D.: A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix, *Econometrica*, 55, 703–708, <https://doi.org/10.2307/1913610>, 1987.
- Nie, W., Kumar, S. V., Chen, J., Zhao, L., Skulovich, O., Yoo, J., Pflug, J., Ahmad, S. K., and Konapala, G.: Rethinking deep learning: linear regression remains a key benchmark in predicting terrestrial water storage, <https://arxiv.org/abs/2510.10799>, 2025.
- Niraula, B. and Goessling, H. F.: Spatial Damped Anomaly Persistence of the Sea Ice Edge as a Benchmark for Dynamical Forecast Systems, *Journal of Geophysical Research: Oceans*, 126, <https://doi.org/10.1029/2021JC017784>, 2021.
- Pechlivanidis, I. G., Crochemore, L., Rosberg, J., and Bosshard, T.: What Are the Key Drivers Controlling the Quality of Seasonal Streamflow Forecasts?, *Water Resources Research*, 56, <https://doi.org/10.1029/2019WR026987>, 2020.
- Reager, J. T., Thomas, B. F., and Famiglietti, J. S.: River basin flood potential inferred using GRACE gravity observations at several months lead time, *Nature Geoscience*, 7, 588–592, <https://doi.org/10.1038/ngeo2203>, 2014.
- Save, H., Bettadpur, S., and Tapley, B. D.: High-resolution CSR GRACE RL05 mascons, *Journal of Geophysical Research: Solid Earth*, 121, 7547–7569, <https://doi.org/10.1002/2016JB013007>, 2016.
- Schreiber, T. and Schmitz, A.: Improved Surrogate Data for Nonlinearity Tests, *Physical Review Letters*, 77, 635–638, <https://doi.org/10.1103/PhysRevLett.77.635>, 1996.
- Shukla, S. and Lettenmaier, D. P.: Seasonal hydrologic prediction in the United States: understanding the role of initial hydrologic conditions and seasonal climate forecast skill, *Hydrology and Earth System Sciences*, 15, 3529–3538, <https://doi.org/10.5194/hess-15-3529-2011>, 2011.
- Springer, A., De Lannoy, G., Rodell, M., Ewerdwalbesloh, Y., Gerdener, H., Khaki, M., Li, B., Li, F., Schumacher, M., Tangdamrongsab, N., Tourian, M. J., Nie, W., and Kusche, J.: A review of current best practices and future directions in assimilating GRACE/-FO terrestrial

- 580 water storage data into numerical models, *Hydrology and Earth System Sciences*, 30, 985–1022, <https://doi.org/10.5194/hess-30-985-2026>, 2026.
- Steidl, V. and Zhu, X. X.: Hierarchical Graph Networks for Forecasting Terrestrial Water Storage Anomalies, in: Machine Learning and the Physical Sciences Workshop, 39th Conference on Neural Information Processing Systems (NeurIPS), <https://ml4physicalsciences.github.io/2025/>, 2025.
- 585 Tapley, B. D., Bettadpur, S., Watkins, M., and Reigber, C.: The gravity recovery and climate experiment: Mission overview and early results, *Geophysical Research Letters*, 31, <https://doi.org/10.1029/2004GL019920>, 2004.
- Wilks, D. S.: “The Stippling Shows Statistically Significant Grid Points”: How Research Results are Routinely Overstated and Overinterpreted, and What to Do about It, *Bulletin of the American Meteorological Society*, 97, 2263–2273, <https://doi.org/10.1175/BAMS-D-15-00267.1>, 2016.
- 590 Wood, A. W. and Lettenmaier, D. P.: An ensemble approach for attribution of hydrologic prediction uncertainty, *Geophysical Research Letters*, 35, <https://doi.org/10.1029/2008GL034648>, 2008.
- Yin, J., Slater, L. J., Khouakhi, A., Yu, L., Liu, P., Li, F., Pokhrel, Y., and Gentile, P.: GTWS-MLrec: global terrestrial water storage reconstruction by machine learning from 1940 to present, *Earth System Science Data*, 15, 5597–5615, <https://doi.org/10.5194/essd-15-5597-2023>, 2023.
- 595 Yossef, N. C., Winsemius, H., Weerts, A., van Beek, R., and Bierkens, M. F. P.: Skill of a global seasonal streamflow forecasting system, relative roles of initial conditions and meteorological forcing, *Water Resources Research*, 49, 4687–4699, <https://doi.org/10.1002/wrcr.20350>, 2013.
- Zhang, L., Shen, Y., Ji, K., Wang, F., and Chen, Q.: An integrated postprocessing approach for extracting time variable signals from GRACE monthly gravity field models, *Journal of Hydrology*, 661, 133 552, <https://doi.org/10.1016/j.jhydrol.2025.133552>, 2025.
- 600 Zhu, E., Yuan, X., and Wood, A. W.: Benchmark decadal forecast skill for terrestrial water storage estimated by an elasticity framework, *Nature Communications*, 10, 1237, <https://doi.org/10.1038/s41467-019-09245-3>, 2019.

Table C1. Conventional metrics for the retained models: pooled RMSE and per-basin median RMSE in centimeters of equivalent water height, and per-basin median correlation (CC) and Nash–Sutcliffe efficiency (NSE) on the anomaly and on the full signal, over 234 basins. The damped-persistence rows use the stronger variant at each lead.

Model	Lead	Pooled RMSE (cm)	Median RMSE (cm)	CC anom.	CC full	NSE anom.	NSE full
Damped persistence	1	5.321	2.74	0.62	0.86	0.33	0.72
	2	6.156	3.32	0.46	0.81	0.09	0.59
	3	6.383	3.57	0.39	0.79	0.02	0.54
	4	6.447	3.65	0.34	0.78	−0.01	0.51
	5	6.689	3.78	0.30	0.77	−0.04	0.48
	6	6.935	3.96	0.23	0.75	−0.08	0.43
Kalman reference	1	5.128	2.66	0.64	0.86	0.36	0.74
	2	5.650	3.23	0.50	0.82	0.18	0.63
	3	5.949	3.46	0.43	0.80	0.06	0.56
	4	6.161	3.57	0.37	0.78	0.00	0.52
	5	6.413	3.72	0.30	0.76	−0.02	0.48
	6	6.690	3.84	0.24	0.74	−0.05	0.44
State ridge	1	5.133	2.66	0.64	0.86	0.36	0.74
	2	5.691	3.26	0.50	0.82	0.17	0.63
	3	5.931	3.46	0.43	0.80	0.06	0.56
	4	6.064	3.55	0.37	0.78	0.01	0.53
	5	6.288	3.68	0.31	0.76	−0.01	0.49
	6	6.603	3.82	0.25	0.74	−0.05	0.45
Flat-12 ridge	1	5.118	2.66	0.64	0.86	0.36	0.74
	2	5.634	3.19	0.51	0.82	0.18	0.63
	3	5.915	3.39	0.45	0.81	0.08	0.57
	4	6.107	3.52	0.37	0.78	0.03	0.53
	5	6.366	3.72	0.30	0.76	−0.02	0.49
	6	6.678	3.85	0.26	0.74	−0.05	0.44
ERA5 flat-12 ridge	1	5.231	2.56	0.69	0.88	0.42	0.76
	2	5.741	3.12	0.55	0.83	0.22	0.65
	3	6.002	3.37	0.48	0.81	0.10	0.58
	4	6.163	3.48	0.39	0.78	0.04	0.54
	5	6.463	3.69	0.33	0.76	−0.02	0.48
	6	6.774	3.87	0.27	0.74	−0.05	0.43